Kirolos Sedra

Wireless Performance & Systems Engineer

Ottawa, Ontario(relocatable between Ottawa, Waterloo)

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Experience

Wireless Sensors and Devices Lab, University of Waterloo **Sep 2024 – Present**
Research Assistant *Full-time, On-site*

- **Industry-Collaborative Project (Apple)**
  * Owned end-to-end design and execution of production-representative wireless validation scenarios across enterprise (Aruba) and consumer APs (ASUS, TP-Link) in controlled anechoic environments, including roaming (802.11r/v, PMK caching, ClientMatch), coexistence, fairness, and resource allocation
  * Designed and automated a large-scale wireless testbed, reducing human intervention by ˜70% via custom orchestration (Ethernet-triggered FTP client, Aruba AirRecorder logging at 1 Hz, automated DNS/DHCP/RADIUS workflows)
  * Resolved critical test-suite incompatibility between two frameworks by adapting and integrating an open-source Objective-C DUT application, restoring full experimental functionality
  * Led advanced performance analysis of Wi-Fi systems, quantifying spectral efficiency across OSI layers and TCP dynamics (slow start, CUBIC, HyStart), and identifying throughput gaps between 160 MHz and 80 MHz configurations
  * Developed RF measurement pipelines using Raspberry Pi and TinySA for calibrated power measurements (50 ohm matching, attenuation control), analyzing channel gain and polarization effects, and generating waterfall plots for frequency–time behavior, enabling validation against digital twin and theoretical models, the system design was subject to space optimization constraints, and it was configured wirelessly using SSH
  * Developed **Python**-based tooling for **automated data collection**, **post-processing**, and **performance visualization** across all wireless measurement campaigns, forming the backbone of the **validation and analysis pipeline**; extended to **real-time system dashboards** for executive-level demonstrations to **Senior Vice Presidents** and market-facing stakeholders
  * Built scalable simulation and experimentation frameworks using MATLAB and NS-3, including a multi-threaded broadcast emulation system preserving throughput across processes; evaluated packet loss under Gaussian noise
  * **BLE Thresholding and Directional Antenna Validation:** Developed and validated a BLE-based engineering solution using nRF52840 DK boards and a fixed-interval BLE beacon device, combining RSSI thresholding and directional antenna behavior to address a real deployment scenario. Solved a real engineering cost-optimization question by proposing the replacement of expensive RF shielding with directional antennas and RSSI thresholding, saving approximately tens of thousands of dollars that would otherwise be spent on shielding in a real engineering deployment. Modified and validated Nordic BLE example firmware and supporting scripts to streamline data collection, progressing through component sanity checks, RSSI measurement campaigns, mathematical threshold derivation, and final threshold validation under practical RF conditions.
- **Industry-Collaborative Project (Rogers, in collaboration with UWaterloo RoboHub)**
  * **End-to-End 5G SA Testbed Validation:** Built, integrated, debugged, and validated a private 5G SA testbed using srsRAN gNB, Open5GS 5GC, USRP B210 SDR, and Quectel RM502Q-AE UE; resolved AMF/SMF/UPF registration and PDU session issues; validated stable UE attachment, PDU session establishment, and user-plane connectivity; used gNB/core logs with AT/QMI modem diagnostics to verify end-to-end SA operation; conducted controlled SNR ladder experiments with attenuators, OTA static setups, and mobility scenarios; investigated UL/DL asymmetry by analyzing MCS/CQI logs from the gNB side; and measured KPIs including throughput, latency under load, RSRP, RSRQ, and SINR
  * Demonstrated the multi-RAT project results across 2 rounds to a panel of approximately 15 Industry Partner attendees as the robot moved through the demonstration trajectories, highlighting storytelling and technical communication skills by using real-time metrics, not only the static metrics already available, to make the results more accessible and compelling for both technical and non-technical stakeholders

∗ **MASc Thesis: Latency-under-Load Characterization of Wi-Fi/5G for Mobile Robot Teleoperation** — conducted rigorous indoor and outdoor Wi-Fi/5G evaluations for mobile robots (single and mesh), with 20 runs per scenario; performed latency-under-load (LuL) characterization by overlaying wireless quality metrics including MCS, SNR, SINR, and RSSI against performance metrics including P99 latency and average throughput; addressed deployment constraints including interference; quantified mobility-induced throughput degradation and LTE anchoring effects; derived novel insights into wireless behaviour under real-world teleoperation conditions; and collected and logged wireless link data via `iw`, `nmcli`, `bluetoothctl`, `tcpdump`, and QMI interface tooling on Quectel chipsets

∗ **4G/5G Walk Test Around City of Waterloo** *(Ongoing):* Aligned timestamped wireless quality logs and performance metrics including throughput and latency under load with GPS data to create cellular street-level heatmaps

∗ **NR-U/Wi-Fi Coexistence Simulation:** Evaluated NR-U coexistence with Wi-Fi using ns-3.35, CTTC 5G-LENA, and NR-U extensions by varying LBT aggressiveness and traffic load; analyzed throughput, latency, jitter, fairness, Wi-Fi starvation, and NR performance penalties under shared 5.2 GHz unlicensed spectrum

Aquasensing **Mar 2026 – Present**
Firmware Specialist *Part-time*

- Led the migration testing and rollout of a new SDK, resolving long-standing technical debt impacting production systems

- **Applied corporate software-engineering best practices learned at Ejada** by introducing disciplined feature ownership, one-by-one committed updates, Git-based version control, structured review habits, and controlled change tracking across firmware and backend workflows
- **Product Owner** for BLE–LTE IoT system, driving firmware roadmap, enforcing Git-based version control practices, coordinating inter-team sprints and test cases through Azure DevOps under Agile methodology, and managing intra-team feature requests, ownership, and tracking using Jira
- **Transferred database safety practices from corporate fintech development into IoT production workflows** by enforcing stored procedures with logging and rollback mechanisms, preventing irreversible database states in production systems
- Organized and supervised comprehensive test suites, ensuring consistency, reproducibility, and alignment with system-level validation goals
- Diagnosed and resolved system-level issues through RF-informed testing, including anechoic chamber experiments to isolate defects and guide firmware improvements
- Spearheaded technical discussions with telecom partners (including Bell IoT), evaluating plans, specifications, and pricing for large-scale IoT deployments
- Directed firmware and system integration efforts for IoT gateways, improving reliability, feature completeness, and production stability
- Led hiring processes for software co-op roles, including technical interviews and candidate evaluation
- Directed system-level validation and QA for IoT gateways, organizing reproducible test suites, diagnosing RF-informed issues through anechoic chamber experiments, enforcing gateway documentation using a Python–SQL tool, introducing an ESP32-based automated test kit for blackout detection and end-to-end sanity checks using GPIO power control, randomized timer pulses, Wi-Fi logging, and POST requests, and creating the Standard Operating Procedure (SOP) for team members to follow when using the automated test kit
- Demonstrated the IoT gateway system to **12Energy Ltd.**, translating technical functionality into business value through professional stakeholder communication, contributing to a commercial deployment of **50 gateways**, and continuing post-sale technical support for deployment, validation, and issue resolution

Aquasensing **Sep 2024 – Feb 2026**
Research Intern (Cloud, IoT & Firmware Systems) *Part-time*

- Developed cloud-integrated IoT applications using React, Django, and Azure for real-time data analytics and system scalability
- Worked with LoRa/LoRaWAN systems to enable reliable device-to-cloud communication
- Developed and debugged firmware for IoT gateway devices, focusing on hardware-software integration and system stability
- Built and automated a test control suite for 70 IoT gateway units, including provisioning workflows using Azure CLI
- Diagnosed cross-layer issues across embedded devices and cloud infrastructure, improving communication reliability and overall system performance

University of Waterloo **Feb 2025 – Apr 2026**
Teaching Assistant *Part-time, Contract Term-Based*

- Teaching Assistant for ECE 106 (Lab, COMSOL), ECE 610 (2 terms, Mininet-based networking project), and ECE 358; responsible for teaching tutorials, proctoring exams, grading assignments/projects, and holding regular office hours

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer *Full-time, Hybrid*

- Engineered and optimized REST APIs and backend microservices for fintech systems, ensuring scalable and reliable service integration while collaborating with a multicultural team based in Saudi Arabia on loan-focused products for Al Rajhi Bank and Emkan Governmental Services
- Architected and developed a new BNPL application for Emkan Saudi Arabia from the ground up using Java Spring Boot, working in a two-person core team with the team lead and handling backend architecture, end-to-end business logic, and system integration
- Improved database query performance by 30% through SQL optimization, indexing, stored-procedure design, and efficient data access patterns
- Increased API performance by 50% by eliminating redundant stored procedures and optimizing data retrieval logic across backend service flows
- Used stored procedures to support API development, database-side debugging, and testing of fintech backend workflows
- Maintained rollback scripts for data and schema changes, reducing deployment risk and supporting controlled recovery during database updates
- Implemented database operation logging inside stored procedures for modification and deletion actions, improving traceability of backend data changes
- Worked across a structured delivery lifecycle including development environments, feature-specific branches, System Integration Testing (SIT), and production environments

Publications

Scalability Analysis of Wi-Fi Broadcast Mechanisms Using Simulation and Hybrid Approaches

Trajectory-Based Performance Characterization of Wi-Fi for Outdoor Mobile Robot Teleoperation

Education

University of Waterloo **Sep 2024 – Aug 2026**
Master of Science in ECE Specialized in Pattern Analysis and Machine Intelligence *Grade: 92% (A+)*

Ain Shams University **Sep 2018 – Jul 2023**
Bachelor of Science in Computer Engineering *CGPA: 3.12/4.00*

- Graduation Project: Digital twin-based self-driving system (**Siemens EDA**) using **Simcenter PreScan**, focused on sensor fusion (RADAR, LiDAR, Camera) and ROS-based integration/testing; Grade: 4.00/4.00

Technical Skills

Programming: Python, C/C++, Java
Wireless Technologies: Wi-Fi 6/6E, 5G NR, LTE, BLE, 802.11k/v/r, RADIUS, PMK Caching
RF & Wireless Testing: SDR (PlutoSDR, USRP), Spectrum Analyzer, Vector Network Analyzer (VNA), Anechoic Chambers, Shielding, Antenna Design & Orientation, Interference & Coexistence Testing
Network Tools & Protocols: Wireshark, ArubaOS, DHCP, DNS, NTP, FTP, HTTPS, TCP/IP, Mininet, SSH
Simulation & Modeling: MATLAB, Simulink, NS-3, GNU Radio, PreScan
Embedded & Systems: ESP32(Wi-Fi),Arduino, STM32, Raspberry Pi, LoRa/LoRaWAN, nRF Connect
Operating Systems: Linux, macOS, Raspberry

Voluntary Involvement

IEEE AP-S Chapter Webmaster and Treasurer. Served as **Media Lead, Director of Service, and Media Co-Director/Director** within a church community. Operated and directed live productions using **OBS**, cameras, microphones, projectors, and presentation systems, while configuring and troubleshooting HDMI, VGA, video, audio, and power connections. Evaluated and tested new media equipment, optimized system reliability and audiovisual performance, and investigated RF interference and link-quality issues affecting wireless HDMI transmission. Developed **standard operating procedures (SOPs)** and delivered hands-on training to onboard new media team members in equipment setup, live switching, troubleshooting, and production workflows. Led a team of 5 in the congregation's **digital transformation** by architecting and developing a centralized **React,**

TypeScript, and Node.js web application adopted congregation-wide. Mentored youth in media and technical leadership, delivered academic instruction to high school students in orphanages, and supported weekly outreach serving underprivileged and homeless communities.

Coursework

University of Waterloo: Data and Knowledge Modelling and Analysis; Image Processing; Computational Intelligence; Text Analytics / NLP.
Ain Shams University: Real-Time Operating Systems using FreeRTOS; Computer Architecture; Software Design Patterns.

Soft Skills

Fluent in English (C1) with working proficiency in French (B1). Strong interest in emerging technologies and developing intuition for improving end-user experience. Presented graduation projects to a management panel and collaborated daily with a multinational team through both verbal and non-verbal communication.